

Challenge (Habanero)

- Q1.** What is the greatest common factor of $12x$ and $18x^2$?
(A) $2x$
(B) $3x$
(C) $6x$
(D) 12
(E) 18
- Q2.** Factor out the greatest common factor from $9y^2 + 12y$
(A) $3y(3y + 4)$
(B) $3y(3y - 4)$
(C) $3y^2(3 + 4)$
(D) $3y^2(3 - 4)$
(E) $3(3y^2 + 4y)$
- Q3.** Simplify the expression $8x^3 + 12x^2$ by factoring
(A) $4x^2(2x + 3)$
(B) $4x(2x^2 + 3)$
(C) $4x(2x^2 - 3)$
(D) $2x(4x^2 + 6)$
(E) $2x(4x^2 - 6)$
- Q4.** In a gaming app, players earn points at a rate of $6x^2 + 8x$ points per level. What is the greatest common factor of the points earned?
(A) $2x$
(B) $2x^2$
(C) $2x(3x + 4)$
(D) $2(3x^2 + 4x)$
(E) $2(3x + 4)$
- Q5.** What is the greatest common factor of $15x^3y$ and $9x^2y^2$?
(A) $3x^2y$
(B) $3xy$
(C) $3xy^2$
(D) $3x^2y^2$
(E) $3x^3y^2$
- Q6.** A tech firm uses $12x^2 + 16x$ bytes of data per user. What is the greatest common factor of the data used?
(A) $4x$
(B) $4x^2$
(C) $4(3x^2 + 4x)$
(D) $4(3x + 4)$
(E) $4x(3x + 4)$
- Q7.** Simplify the expression $20x^2y + 12xy^2$ by factoring
(A) $4xy(5x + 3y)$
(B) $4xy(5x - 3y)$
(C) $4x(5xy + 3y^2)$
(D) $4x(5xy - 3y^2)$
(E) $2xy(10x + 6y)$
- Q8.** A scoreboard displays $9x^2 + 12x + 3$ points for each player. What is the greatest common factor of the points displayed?
(A) 3
(B) $3x$
(C) $3x^2$
(D) $3(3x^2 + 4x + 1)$
(E) $3(3x + 4)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression $24x^3y^2 + 18x^2y^3$ and simplify. Show all steps.
- Q10.** A coding app uses $15x^2y + 10xy^2$ bits of data per user. Factor out the greatest common factor and explain its significance in the context of the app. Show all steps.

Chef's Tips

- Emphasize the importance of identifying the greatest common factor in simplifying expressions
- Use real-world examples, such as tech and gaming, to illustrate the application of factoring
- Encourage students to check their work by multiplying the factored expression